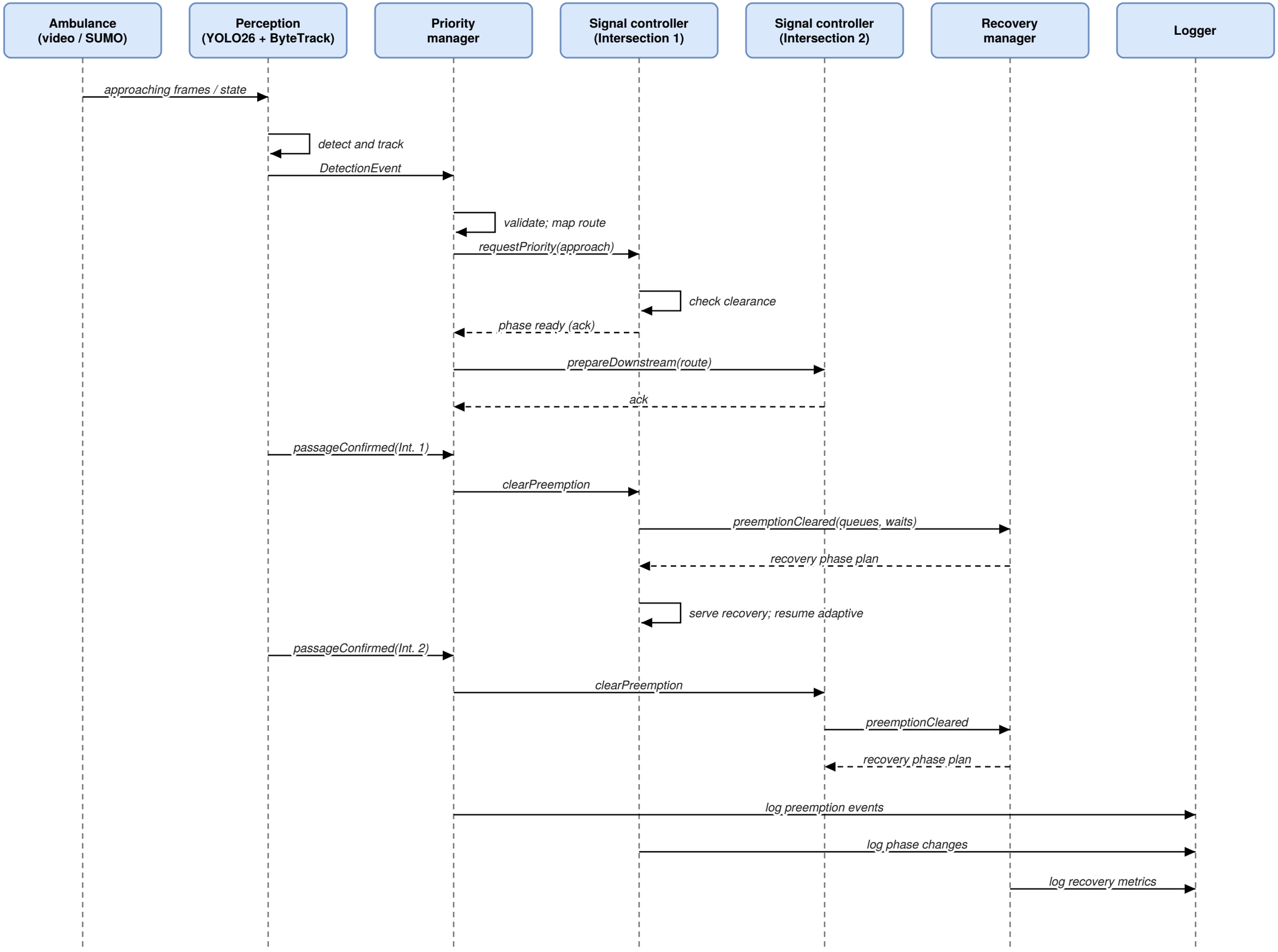

Ambulance (video / SUMO)
Perception (YOLO26 + ByteTrack)
Priority manager
Signal controller (Intersection 1)
Signal controller (Intersection 2)
Recovery manager
Logger
approaching frames / state
detect and track
DetectionEvent
validate; map route
requestPriority(approach)
check clearance
phase ready (ack)
prepareDownstream(route)
ack
passageConfirmed(Int. 1)
clearPreemption
preemptionCleared(queues, waits)
recovery phase plan
serve recovery; resume adaptive
passageConfirmed(Int. 2)
clearPreemption
preemptionCleared
recovery phase plan
log preemption events
log phase changes
log recovery metrics